BRAC University
Department of Computer Science and Engineering

Examination: Quiz 01 Semester : Spring26
Duration: 20 Minutes Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:12

1. a.This line is a preprocessor directive. In the C compilation process, preprocessing happens before lexical analysis. The preprocessor handles lines beginning with #.

   So when the compiler starts processing the code:

   - ➔ The preprocessor runs first.
   - ➔ It processes #include <stdio.h>.
   - ➔ It replaces it with the contents of the header file (or prepares it for compilation).

   After preprocessing, the lexical analyzer receives the expanded source code, not the original directive. Because of this, the lexical analyzer never tokenizes the original #include lines.

   b. Even though the lexical analyzer discards or ignores #include lines, the compiler still keeps track of them by recording include directives during preprocessing. This allows the compiler and development tools to know that an include occurred.

2. A context-free grammar (CFG) that is ambiguous cannot be reliably used in the syntax analysis phase because an ambiguous grammar allows more than one valid parse tree for the same input string, which prevents the parser from determining a unique syntactic structure. A compiler must assign one precise structure so that semantic analysis, type checking, and code generation can proceed deterministically. Ambiguity destroys this requirement.

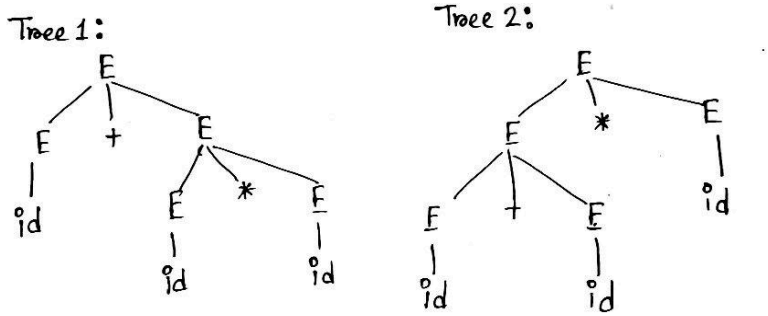
   Example:

   $E \rightarrow E + E$

   $E \rightarrow E * E$

   $E \rightarrow id$

   Input String: id + id * id



   Thus, the same input string has two different interpretations.